

Table 1: Examples of a Tautology and a Contradiction

P	$\neg P$	$P \vee \neg P$	$P \wedge \neg P$
T	F	T	F
F	T	T	F

1.3.2 Logical Equivalences

Definition 2: The compound propositions P and Q are called logically equivalent if $P \leftrightarrow Q$ is a tautology. The relation $P \equiv Q$ denotes that P and Q are logically equivalent.

One way to determine whether two compound propositions are equivalent is to use a truth table. In particular, the compound propositions P and Q are equivalent if and only if the columns giving their truth values agree.

Table 2: De Morgan's Laws

$$\neg(P \wedge Q) \equiv \neg P \vee \neg Q$$

$$\neg(P \vee Q) \equiv \neg P \wedge \neg Q$$

Example 2: Show that $\neg(P \vee Q)$ and $\neg P \wedge \neg Q$ are logically equivalent.

Table 3 Truth Tables for $\neg(P \vee Q)$ and $\neg P \wedge \neg Q$

P	Q	$(P \vee Q)$	$\neg(P \vee Q)$	$\neg P$	$\neg Q$	$\neg P \wedge \neg Q$
T	T	T	F	F	F	F
T	F	T	F	F	T	F
F	T	T	F	T	F	F
F	F	F	T	T	T	T

Example 3: Show that $P \rightarrow \neg Q$ and $\neg P \vee \neg Q$ are logically equivalent. (This is known as the conditional-disjunction equivalence.)

Table 4 Truth Tables for $\neg P \vee \neg Q$ and $P \rightarrow \neg Q$

P	Q	$\neg P$	$\neg P \vee \neg Q$	$P \rightarrow \neg Q$
T	T	F	F	F
T	F	F	T	T
F	T	T	T	T
F	F	T	T	T